JCTC Journal of Chemical Theory and Computation

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All-Atom Reactive Monte Carlo Molecular Dynamics for Molecular Doping in Organic Semiconductors

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Cite This: https://doi.org/10.1021/acs.jctc.5c01206 Read Online

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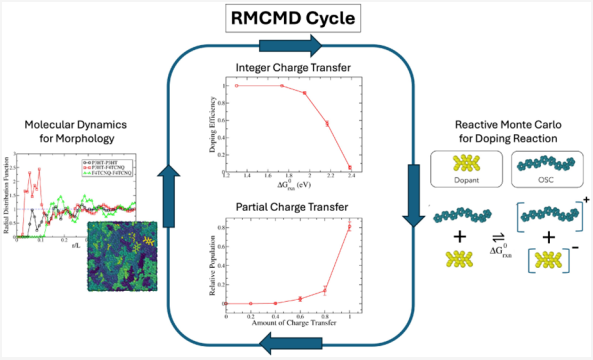
ABSTRACT: The computational study of molecular doping in organic semiconductors (OSCs) is challenged by multiple competing length and time scales. We present an all-atom Reactive Monte Carlo Molecular Dynamics (RMCMD) method for quantitatively determining molecular doping efficiency in OSCs. A Metropolis criterion is employed for the doping reaction, which is parametrized from density functional theory (DFT) calculations of energetics and atomic partial charges of the doped and neutral molecular species. Polaronic effects are included in the RMCMD method to enable geometric reorganization upon doping, including the flattening of the inter-ring dihedrals. Extensions to partial charge transfer states are also developed to allow for the inclusion of charge transfer complexes. To demonstrate the validity of the approach, the doping efficiency and radial distribution function of a poly(3-hexylthiophene) system doped with 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane are calculated in an amorphous morphology. The method is implemented as a fix in LAMMPS, with the code made publicly available on GitHub.



1. INTRODUCTION

Organic semiconductors (OSCs) are molecules or polymers with conjugated π-electron systems that facilitate semiconducting behavior, enabling a range of technological applications including organic light-emitting diodes (OLEDs)[1−3] and organic thin-film transistors (OTFTs). OSCs can be solution or vapor deposited on plastic substrates at low temperatures, allowing for the creation of flexible large-area displays and wearable technologies.[4,5] However, the primary drawback of the reduced processing costs and scalability of OSCs is that most OSCs suffer from significantly lower electronic conductivities than their inorganic counterparts.[6,7] These low conductivities are typically ascribed to a combination of chemical and morphological disorder in OSCs: weak intermolecular bonding and irregular π-π stacking result in partially ordered or disordered morphologies that can lead to electron localization and low densities of available charge carriers for conduction.[8−11] The injection of additional charge carriers through molecular doping is a ubiquitous approach for boosting the conductivity of OSCs.[12]

Understanding the design principles of molecular doping in OSCs is challenged by the strong coupling of electronic structure and morphology. Electronically, there are two main types of electronic states that govern the doping process between molecular dopants and OSCs, with poly(3-hexylthiophene) (P3HT) doped with 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ) being the most studied OSC-dopant combination.[13−18] The Integer Charge Transfer (ICT) model treats the doping reaction as the transfer of a charge carrier (either hole or electron) from a dopant to an OSC. In contrast, the Charge Transfer Complex (CTC) model treats the doping reaction as the formation of a hybridized molecular orbital between dopant and OSC, resulting in partial charge transfer between the OSC and dopant molecules.[19,20] An additional complication is that while most dopants are capable of forming ICTs, only a small proportion of these integer charge carriers participate in conduction.[20] As a result, OSCs can require doping concentrations as high as 20%,[21,22] in contrast to conventional inorganic semiconductors that require no more than 10^{18} cm^{-3} ($\approx$ 0.01%) doping concentration.[23] When doped at such high concentrations, F4TCNQ can enter crystalline regions of P3HT and disrupt the π-π stacking, affecting electron delocalization. Mixed solution-processed P3HT:F4TCNQ show dopant-induced aggregation, which results in significantly lower conductivities.[24] Interestingly, the reverse influence is also possible, in which polaron formation in amorphous doped P3HT can cause significant morphological changes.[25,26] Available experimental evidence to-date shows that doping behavior is strongly coupled to the

Received: July 22, 2025
Revised: September 29, 2025
Accepted: September 30, 2025

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morphology evolution in doped OSCs, and hence an ideal theoretical treatment should treat this coupling directly.

Molecular modeling has shed light on the fundamental processes governing molecular doping.[27,28] DFT calculations have demonstrated a linear relationship between the difference between the ionization potential (IP) of an OSC molecule and the electron affinity (EA) of a dopant molecule (for p-type OSCs), IP-|EA|, and the amount of charge transfer between an OSC-dopant dimer.[29] All-atom molecular dynamics (MD) simulations on bulk P3HT:F4TCNQ system combined with thermodynamic integration showed that the overall free energies of forming the fully doped ICT and CTC polymorphs are similar, explaining why both phases always appear to be present experimentally.[30] Excited state GW-BSE calculations on oligothiophene:F4TCNQ stacks showed that lengthening the donor backbone reduced the CTC band gap by several hundred meV.[31] First-principles work examining multimolecule effects found that adding an extra dopant or OSC molecule to the aggregate can significantly lower the barrier to charge transfer.[32] Short range overscreening can occur in doped OSCs, where the generated hole moves in a flatter electrostatic potential as opposed to the deep Coulomb trap at the host cation.[33] It has been shown that the occurrence of an ICT state in PCPDT-BT copolymers depends on the local polymer conformation surrounding the dopant molecule.[34] The size of the counterion *i.e.*, the ionized dopant can also play a significant role in the electronic and thermoelectric properties of the doped OSC.[35] These studies reinforce the critical importance of molecular modeling in extracting design principles for molecular doping and also emphasize the importance of multimolecule effects in realistic morphological environments in controlling the charge transfer process.

Provided the importance of treating the strong coupling of electronic structure and morphology in complex, heterogeneous environments, a computational treatment of doping efficiency (defined as the fraction of ionized dopants) as a function of morphology is a challenging endeavor. *Ab-initio* molecular dynamics (AIMD) can, in principle, capture both electronic and conformational properties. However, this approach is infeasible for systems with more than a few hundred atoms or for time scales beyond a nanosecond. Classical MD with carefully tuned interatomic potentials can yield accurate equilibrium morphologies for systems with millions of atoms. However, classical MD does not shed any light on the electronic behavior of the system. The Reactive Monte Carlo (RMC) method[36−39] implemented in the semigrand canonical ensemble,[40] represents an interesting pathway to a solution as a powerful technique capable of modeling the equilibrium behavior of nonideal chemically reacting systems. Since its proposal in 1994, the method has been applied to a wide variety of complex and interesting systems.[41] Mullen et al.[42] used RMC simulations to study CO_2 absorption in triethyl(octyl)phosphonium 2-cyanopyrrolide, an anion ionic liquid. Mullen and Maginn[43] also applied RMC to study xylene isomerization in vapor, liquid, and supercritical phases, obtaining good agreement with experiment. Leiding and Coe[44] applied *ab-initio* Reactive Monte Carlo to NH_3, and observed regions of state space for which the RMC sampling is much more efficient than AIMD. Fetisov et al.[45] also applied a first-principles based Reactive Monte Carlo method to study a nitrogen−oxygen mixture at conditions that are present in atmospheric lightning strikes and explosions. The replica exchange method has also been adapted in the framework of the RMC method, allowing for more efficient simulations.[46] These examples are a testament to the efficacy and versatility of the RMC method for treating reactivity in complex molecular environments.

Inspired by these developments, Verma and Jackson introduced the RMC approach for modeling molecularly doped OSCs.[47] This is a simple MC method that uses swap and displacement moves for morphology equilibration, and a reactive MC move[48] to model the doping reaction. This method was applied to P3HT doped with F4TCNQ and successfully captured the qualitative effects of dopant aggregation on the doping efficiency. However, the method is fully classical and represents the OSC and dopant molecules as coarse-grained beads, which has obvious limitations for treating critical elements such as molecular packing, local electrostatics, and polaronic effects. Here, we develop a fully atomistic Reactive Monte Carlo Molecular Dynamics (RMCMD) approach for the doped OSC system, based on Verma and Jackson's RMC method. The RMCMD method is a novel and powerful approach for modeling the doping behavior in OSCs, and is capable of capturing the interplay between the formation of charge transfer states and morphological evolution at the characteristic length and time scales of soft materials. To enable widespread application of our method, it is implemented as a fix in LAMMPS[49,50] and made publicly available on GitHub.[51]

The paper is organized as follows. First, we detail the RMCMD method and the workflow employed that leverages DFT parametrization. We apply our method to the fruit-fly of the molecular doping community: P3HT doped with F4TCNQ. We first demonstrate the approach incorporating only ICT states, and then progress to also include a spectrum of CTC states as well as polaron induced backbone planarization in the doped state. In all cases, we obtain qualitative agreement with previous experimental results and prior first-principles calculations on single molecule and dimer systems,[17,20,29] Finally, we highlight future areas of improvement for the method as well as future applications of the RMCMD method to molecular doping.

2. REACTIVE MONTE CARLO

Consider a binary molecular system consisting of two molecule types: OSCs and molecular dopants. The present development focuses on *p*-type dopants in which an electron is transferred from the OSC to the dopant, and holes on the OSC are the operative charge carriers—an identical approach applies to *n*-type dopants. The simplest treatment of doping reactions considers the case of ICT of an electron from an OSC to a dopant, the reaction of which can be expressed as eq 1

$$\mathrm{OSC} + \mathrm{Dopant} \overset{\Delta G_{rxn}}{\rightleftharpoons} [\mathrm{OSC}]^+ + [\mathrm{Dopant}]^- \tag{1}$$

and is characterized by the free-energy of the reaction ΔG_{rxn}. Within the simulation box we model this reaction as a RMC step[42−44,47,48] where either (i) a neutral OSC and a neutral dopant molecule are randomly selected and replaced with their charged counterparts, or (ii) a positively charged OSC and a negatively charged dopant are randomly selected and replaced with their neutral counterparts. A Metropolis-style acceptance criterion applies for the forward and reverse reactions, consistent with the semigrand ensemble, that can be derived from the grand canonical partition function (eq 2)

https://doi.org/10.1021/acs.jctc.5c01206
J. Chem. Theory Comput. XXXX, XXX, XXX−XXX

$$p = \min\left(1, \frac{N_{OSC}N_{Dopant}}{([N_{OSC}]_+ + 1)([N_{Dopant}]_- + 1)} \exp(-\beta\Delta G_{rxn})\right) \quad (2)$$

where N_{OSC}, N_{Dopant} refer to the number of neutral OSC and dopant molecules, $[N_{OSC}]_+$,$[N_{Dopant}]_-$ refer to the number of charged OSC and dopant molecules, and ΔG_{rxn} represents the free-energy of the ionization/recombination reaction between the OSC and dopant. The doping efficiency is then given by $[N_{Dopant}]_-/N_{Dopant}$. Here, we choose to approximate the full quantum mechanical energy of the ionization/recombination reaction in the multimolecule system (eq 3) using a classical force-field representation of the potential energy surface (PES) supplemented by an electronic energy difference between the minima of the neutral and doped PES's (ΔU_{elec}). In eq 3, ΔU_{ES} represents the change in the classical electrostatic energy of interaction of the system, obtained using atomic partial charges, ΔU_{intra} represents the change in potential energy due to the change in the curvature of intramolecular PES's upon doping, and ΔU_{LJ} represents the change in intermolecular interaction energy of the system resulting from difference in the atomic Lennard-Jones parameters between the neutral and doped states. ΔU_{LJ} is assumed to be zero throughout this work, and hence dropped in future equations.

$$\Delta G_{rxn} = \Delta U_{elec} + \Delta U_{LJ} + \Delta U_{intra} + \Delta U_{ES} \quad (3)$$

In this work, ΔU_{elec} is represented in terms of the single-molecule IP and EA of the OSC and Dopant (eq 4).

$$\Delta G_{rxn} = (IP_{OSC} - EA_{Dopant} + \Delta U_{intra}) + \Delta U_{ES} \quad (4)$$

For simplicity, we combine ΔU_{elec} and ΔU_{intra} into a single term, denoted ΔG^0_{rxn}, which represents all single molecule energetic terms contributing to the reaction free energy change, enabling parameter sweeps within our model as a function of the single parameter, ΔG^0_{rxn}, representing the exothermicity of the doping reaction.

$$\Delta G_{rxn} = \Delta G^0_{rxn} + \Delta U_{ES} \quad (5)$$

In prior work that introduced the RMC method for molecular doping, OSCs and dopants were represented simplistically by single coarse-grained beads carrying a charge, q, denoting the neutral or doped charge states of the molecules. With this prior simplification, switching from neutral to charged species simply involved setting the charge of each bead to ±1, and evaluating ΔU_{ES} using a short-range electrostatics and an Ewald summation.[52] However, in the all-atom representation, the atomic partial charges within each OSC and dopant molecule need to be switched to their associated values defined in the neutral or doped states of the molecules. To obtain the atomic partial charges corresponding to the neutral and doped states of the OSC and dopant molecules, we use Löwdin[53] atomic partial charges extracted from geometry optimization of the OSC and dopant molecules, respectively, at their specified integer charge states. DFT calculations are performed at the ωB97X-D3[54,55] /def2-SVP[56,57] level of theory in Orca.[58] Note that any method for atomic partial charge determination can be used for this task, but here Löwdin[53] charges are selected for simplicity.

An important point relates to how the electrostatic energies calculated from the atomic partial charges are incorporated into the energetic criteria of eq 5. If not done carefully, there is the potential for double counting the intramolecular electrostatic contribution evaluated at the single molecule DFT level as well as that associated with the intramolecular electrostatic contributions evaluated via the atomic point charges. In evaluating this term, we assume that the intramolecular electrostatic terms are most accurately accounted for within the IP and EA estimates of the single molecules. When considering the Ewald-style summation[52] of short-range and long-range Coulombic interactions, we use the decomposition of the electrostatic energy in eq 6.

$$\Delta U_{ES} = \Delta U^{SR}_{ES} + \Delta U^{LR}_{ES} - \Delta U^{intra-SR-OSC}_{ES} - \Delta U^{intra-SR-Dopant}_{ES} \quad (6)$$

where ΔU^{SR}_{ES} is the short-range, real-space contribution to the electrostatic energy, ΔU^{LR}_{ES} is the long-range, Fourier-space contribution to the electrostatic energy, and $\Delta U^{intra-SR-OSC}_{ES}$ and $\Delta U^{intra-SR-Dopant}_{ES}$ are the intramolecular electrostatic energy change of an OSC and Dopant molecule, respectively. Note that the evaluation of the eq 6 quantities exhibits trivial overhead cost as the calculation of the intramolecular electrostatic energy change within the OSC and Dopant must be evaluated and subtracted off each electrostatic energy evaluation in LAMMPS.[49,50] The average values of $\Delta U^{intra-SR-OSC}_{ES}$ and $\Delta U^{intra-SR-Dopant}_{ES}$ are 0.97 and 0.7 eV, respectively, with full distributions provided in Figure S1.

3. MOLECULAR DYNAMICS SIMULATIONS

Whereas in previous work random displacement and identity swap MC moves were employed in concert with the RMC moves to relax the system, we have elected to employ a hybrid methodology using MD simulations to relax the configurations of the system. This approach benefits from the high efficiency and parallelization of LAMMPS[49,50] as well as the potential additional flexibility and customizability of the MD methods for future use. Our implementation follows the standard format of hybrid MC/MD methods in which an alternation between a certain number of reactive MC moves and a certain number of MD steps are conducted. In the simulations that follow we employ a time step of 1 fs and periodic boundary conditions. The Lennard-Jones cutoff was chosen to be 12 Å, with geometric mixing for the unlike atom interactions. We employ Ewald summation with an accuracy of 1e-5 and the cutoff for short-range electrostatics was chosen to be 20 Å. The P3HT 8-mers as well as F4TCNQ were modeled using the OPLS-AA force field.[59,60] The LAMMPS input, geometry and output files for the runs in this paper are made available on GitHub[51] to aid replication.

3.1. Charge Transfer Complex. As there is substantial evidence for the formation of partial charge transfer states in doped systems via CTC states[19,20] we have adapted the above methodology to be able to incorporate an arbitrary number of partial charge transfer states into the RMC simulation. We include this effect in our RMC method by considering a discrete set of partial charge states between 0 and ±1. For n charge states (q_1, q_2,q_3...,q_n) with $q_1 = 0$ and $q_n = 1$, we allow RMC steps between any two of the states. This can be represented as

https://doi.org/10.1021/acs.jctc.5c01206
J. Chem. Theory Comput. XXXX, XXX, XXX−XXX

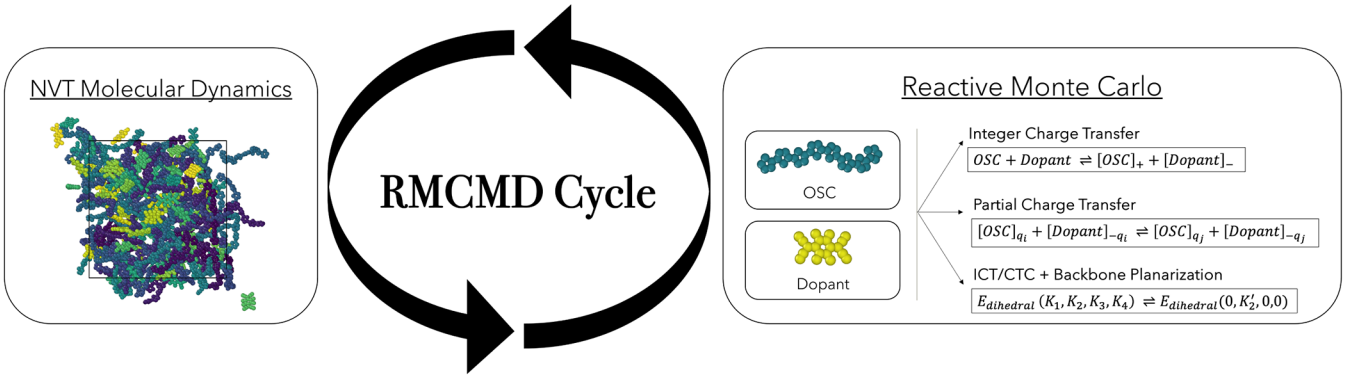

Figure 1. Schematic depicting the RMCMD algorithm, alternating between cycles of NVT Molecular Dynamics and Reactive Monte Carlo moves. Three types of reactive moves (integer charge move, partial charge transfer, and dihedral flattening) are available in our implementation.

$$[\mathrm{OSC}]_{q_i} + [\mathrm{Dopant}]_{-q_i} \rightleftharpoons [\mathrm{OSC}]_{q_j} + [\mathrm{Dopant}]_{-q_j} \tag{7}$$

where the forward reaction is achieved by picking an OSC-Dopant pair with partial charge $\pm q_i$ and switching the charge to $\pm q_j$. The acceptance criteria for the reaction is

$$p = \min\left(1, \frac{[N_{\mathrm{OSC}}]_{q_i}[N_{\mathrm{Dopant}}]_{-q_i}}{([N_{\mathrm{OSC}}]_{q_j} + 1)([N_{\mathrm{Dopant}}]_{-q_j} + 1)} \exp(-\beta\Delta G_{\mathrm{rxn}})\right) \tag{8}$$

While the approach in principle follows the same procedure as for ICT states defined previously, it is difficult in practice to estimate ΔG_{rxn} and the associated atomic partial charges for partial charge states. While in principle constrained DFT could be used to compute these quantities directly, here we adopt a simpler approach. To obtain estimates of the atomic partial charges, we use a linear interpolation between the atomic partial charges in the neutral and ICT states, though this simple ansatz will need to be further addressed in future work. Estimates of ΔG_{rxn} for partial charge states are obtained by fitting DFT-derived partial charge transfer data from our previous work[61] against the associated IP and EA quantities from those same calculations (Supporting Information). Specifically, this is done by obtaining the total ΔG vs the amount of OSC-Dopant charge transfer from DFT, and subtracting the classical electrostatic energy utilizing the atomic partial charges to get the purely "quantum mechanical" portion of the free energy change for use in the RMC simulations.

3.2. Incorporation of Polaronic Effects. It is well established that the dominant charge carriers in p-type OSCs are positively charged polarons, quasiparticles that represent the interaction between holes and their local nuclear environments via vibrations. Polaron formation can have significant effects on the behavior of the doped OSC, exacerbating charge localization in disordered P3HT. With a primary goal of the RMCMD methodology developed here being to capture the interplay of morphology and electronic state changes in molecularly doped OSCs, it is critical to capture the geometric changes of the OSC and dopant induced by the change in electronic state after the doping reaction. Perhaps the most important polaron-induced geometrical change in OSCs is the stiffening of the conjugated OSC backbone mediated by planarization of the conjugated inter-ring dihedrals. Here, we explicitly incorporate this planarization effect within our model by modifying the force-field parameters of OSCs upon being switched from the neutral to the doped electronic state. In practice, this is done by switching the relevant set of conjugated inter-ring dihedral coefficients to favor a more planar structure. The OPLS form of the dihedral potential used in this work takes the form of eq 9.

$$E = K_1(1 + \cos\phi) + K_2(1 - \cos 2\phi) + K_3(1 + \cos 3\phi) + K_4(1 - \cos 4\phi) \tag{9}$$

Our aim is to find coefficients K_1 to K_4 such that the E is minimized when $\phi = 0$ or $\phi = \pi$. If K_1, K_3 and K_4 is set to 0, the energy becomes

$$E = K_2(1 - \cos 2\phi) \tag{10}$$

For $K_2 > 0$, E is minimized when $(1 - \cos 2\phi) = 0$ i.e., when $\phi = 0$ or π. Ideally, the value of the dihedral coefficient K_2 for the polaronic (doped) state should be obtained from first-principles calculations. The magnitude of K_2 determines the strength of the dihedral forces that drive flattening, i.e.

$$F = -\frac{\partial E}{\partial \phi} = 2K_2 \sin 2\phi \tag{11}$$

For this work, since we are primarily interested in studying the qualitative effects of polaron-induced dihedral flattening on the morphology, we select a sufficiently large arbitrary value of K_2 ($\approx$ 0.87 eV) to ensure the flattening of the dihedral in the doped polaronic state. As long as planarization is achieved, any sufficiently large value of K_2 can be chosen. If the dihedral coefficients are switched from (K_1', K_2', K_3', K_4') to $(0,K_2,0,0)$, there is a corresponding energy change

$$\Delta E = -K_1'(1 + \cos\phi) + (K_2 - K_2')(1 - \cos 2\phi) - K_3'(1 + \cos 3\phi) - K_4'(1 - \cos 4\phi) \tag{12}$$

In principle, this polaronic energy change should also be incorporated as an additional term in eq 3. For simplicity, we lump this change in energy directly into the catch-all term $\Delta G^0_{\mathrm{rxn}}$. In Figure S3, we provide histograms quantifying the distribution of this polaronic energy contribution across the sampled configurational space. This approach can also be trivially extended to include changes in the bond and angle potentials between the neutral and doped states, but such

https://doi.org/10.1021/acs.jctc.5c01206
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effects are not incorporated here due to (i) the time and computational cost of parametrizing these potential energy surfaces and (ii) the expectation that these small changes in geometry will play a minimal role relative to the dihedral flattening. More rigorous and quantitative treatments of polaronic effects within this approach are left to future work.

3.3. Workflow and Implementation. Provided the details of the implemented RMCMD algorithm, we apply this algorithm to examine doping in the fruit-fly system of P3HT and F4TCNQ. Figure 1 shows the overall workflow that we employ in this work. NPT simulations are run on a box with 100 P3HT molecules (8mers, 202 atoms per molecule) and 25 F4TCNQ molecules (20 atoms per molecule) until a density of 1−1.1 g/cm^3 is achieved. We then alternate between 1 ps of NVT MD and 200 RMC moves, for 10 ns. We run three sets of simulations. The first involves only ICT reactions, performed at different values of ΔG_{rxn}. The second set allows for both ICT and CTC type reactions, using a single ΔG_{rxn} profile obtained from DFT. The final set of calculations are ICT only, but allow for dynamic dihedral planarization upon polaron formation. For each set of simulations, we examine the doping efficiency and the radial distribution function to quantify the electronic and morphological states that form upon doping.

The RMCMD algorithm is implemented as a fix in LAMMPS and made publicly available on GitHub.[51] As is required for LAMMPS fixes, it supports parallelism, but the Monte Carlo moves are performed sequentially. While this work is focused on P3HT:F4TCNQ, the implementation allows for the study of any OSC-Dopant pair for which a suitable classical force-field exists.

4. RESULTS

First, we perform a simple calibration of our RMC implementation. We prepare a system with equal numbers of P3HT (represents here as 8-mers) and F4TCNQ molecules within the simulation box and set the dielectric constant to be very large ($\epsilon = 10^{11}$) and very small ($\epsilon = 0.05$). The resulting doping efficiencies using the simulation parameters described in the Workflow and Implementation section are shown in Figure 2, calculated as a function of ΔG^0_{rxn}. At the high dielectric limit, the electrostatics are effectively nonexistent, making the doping efficiency of the system determined solely by the reaction free energy, ΔG^0_{rxn}, and the OSC and dopant concentrations. For an equal number of OSC and dopant molecules, and $\Delta G^0_{rxn} = 0$, then the doping efficiency is determined solely by entropy and the doping efficiency should be 50%. At the unphysically low dielectric limit, the electrostatic interaction energy is multiple orders of magnitude larger than the thermal energy, and the observed doping efficiency should be insensitive to ΔG^0_{rxn}. The sensible behavior of the RMCMD methodology in both limits validates the RMC implementation and corroborates the findings of our previous work utilizing a simple Lennard-Jones fluid.

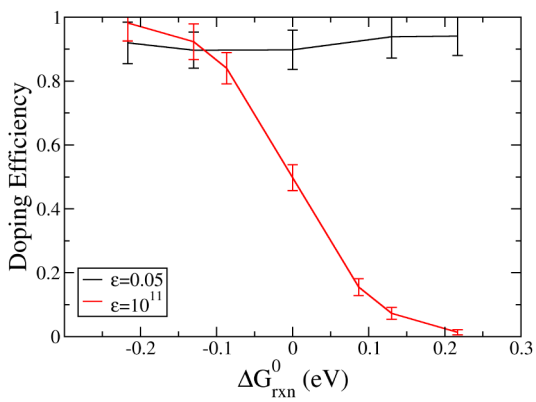

Figure 2. Doping efficiency as a function of the reaction free energy, ΔG^0_{rxn}, for an amorphous system with 100 P3HT and 100 F4TCNQ molecules at high dielectric (red) and low dielectric (black) limit.

Provided the validation of the RMCMD model, we next explore how the doping efficiency in amorphous morphologies of P3HT/F4TCNQ is affected by varying the value of ΔG^0_{rxn}. Of course, in the experimental system ΔG^0_{rxn} would exhibit a narrow peak around a single value corresponding to the intrinsic $IP - EA$ and polaron energetics of P3HT/F4TCNQ, but here tuning ΔG^0_{rxn} provides a knob for examining how the electronic free-energy difference affects the molecular doping efficiency. We selected a concentration of F4TCNQ of 25% which represents the upper limit of what is experimentally attainable, but is employed to improve the statistics of the RMC sampling procedure. Figure 3 shows the calculated doping efficiency as a function of ΔG^0_{rxn} for the real dielectric of the system of 1. We observe the calculated doping efficiency as a function of ΔG^0_{rxn} spans from fully ionized to no ionization over an energy range of 1.4−2.4 eV. Note that doping here occurs despite the endothermic ΔG^0_{rxn} value due to the Coulombic energy between positive and negative charge in the doped state, that here is being explicitly included in the energetic evaluation via the short-range electrostatics and long-range Ewald summation. Importantly, the observed range of ΔG^0_{rxn} values over which a broad range in doping efficiency is observed qualitatively matches the energy ranges observed from absorption spectra studies of doped P3HT[17,20] as well as the first-principles studies of charge transfer between P3HT-F4TCNQ pairs,[29] providing additional validation of the RMCMD methodology.

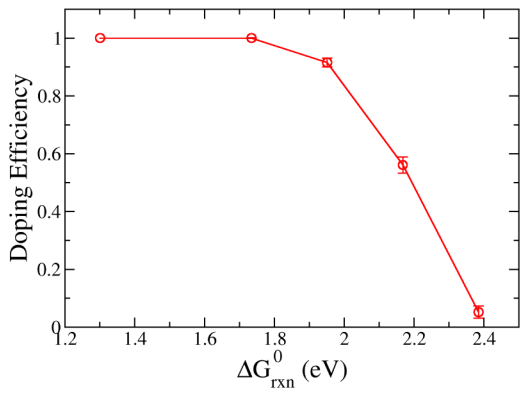

Figure 3. Doping efficiency as a function of the reaction free energy, ΔG^0_{rxn}, for the amorphous system with 100 P3HT and 25 F4TCNQ molecules simulated at a dielectric of 1.

One of the main motivations for the RMCMD method is not simply to determine how the formation of doped electronic states depends on the underlying morphology, but

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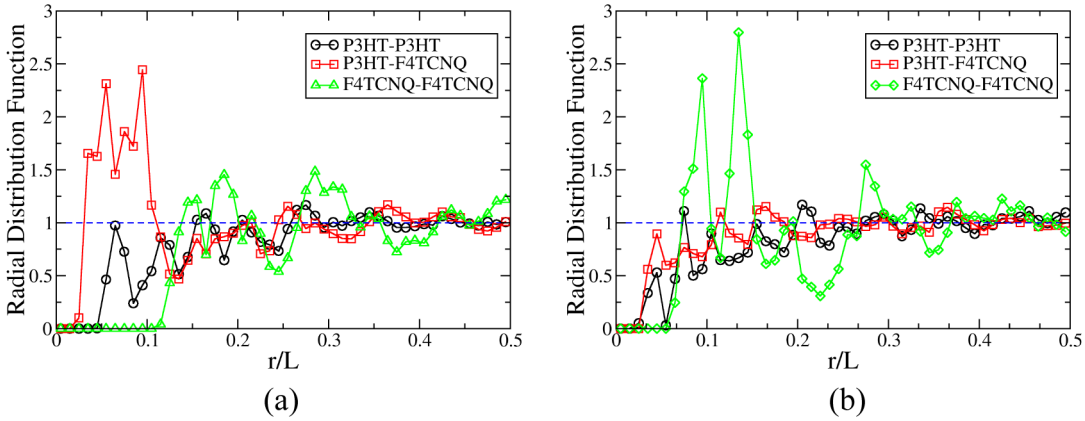

Figure 4. Radial distribution functions for an amorphous system of 100 P3HT and 25 F4TCNQ molecules at (a) ΔG_{rxn} = 1.3 eV (b) ΔG_{rxn} = 2.4 eV and dielectric equal to 1. Here L refers to periodic cell size.

also to understand how, upon doping, the morphology itself evolves. By alternating between RMC reaction moves and configurational relaxation via MD, the simulation box is allowed to adjust its molecular packing to accommodate the changes in electronic states. Figure 4 shows the RDF for the amorphous, dielectric equal to 1 system at $\Delta G^0_{rxn} = 1.3$ eV (Figure 4a) and $\Delta G^0_{rxn} = 2.38$ eV (Figure 4b). At $\Delta G^0_{rxn} = 1.3$ eV, all dopants are ionized, and the electrostatic interactions resulting from the large population of doped species result in shorter intermolecular distances between ionized P3HT-F4TCNQ pairs and larger intermolecular distances between pairs of F4TCNQ molecules and pairs of P3HT molecules (Figure 4a). In contrast, at $\Delta G^0_{rxn} \approx 2.4$ eV, no doping reactions occur, and the morphology is dictated by the standard interatomic potentials of P3HT and F4TCNQ instead of the electrostatic interactions of the doped species. This morphological segregation validates the ability of the RMCMD model to capture doping-induced changes in the morphology, which is an important enabling feature of this new methodology relative to existing simulation approaches for OSCs.

Provided the validation of RMCMD to capture molecular doping behavior in amorphous P3HT/F4TCNQ using ICT state representations of the doped system, we next explore the ability of the method to accommodate the inclusion of CTCs using partial charge RMC moves. However, studying partial charge transfer in P3HT/F4TCNQ requires ΔG^0_{rxn} as a function of the amount of charge transfer in the CTC states. For the case of RMCMD with only ICT states, a sweep over ΔG_{rxn} between the neutral and ICT states was practical, but such a strategy would clearly be impractical provided the many different permissible charge states. To parametrize the energies of the different possible CTC states, we leverage prior first-principles calculations[61] of partial charge transfer in all-atom representations of P3HT/F4TCNQ, the results of which are summarized in Figure S2. Atomistic MD simulations of amorphous P3HT/F4TCNQ morphologies are performed and neighboring P3HT/F4TCNQ dimers are extracted and DFT-optimized to determine the amount of ground state charge transfer using Natural Population Analysis and the total energy of the system. A linear trend is observed between the amount of charge transfer in the CTC state and the total DFT energy change relative to the neutral baseline (ΔG_{DFT}), so a linear fit to this form can be extracted. We can then express the doping free-energy change in the system as the difference between the DFT free-energy change, ΔG_{DFT}, and the electrostatic energy change, ΔU_{ES}.

$$\Delta G^0_{rxn} = \Delta G_{DFT} - \Delta U_{ES} \tag{13}$$

Here, we consider six evenly spaced partial charge states −0, ±0.2, ±0.4, ±0.6, ±0.8, and ±1.

ΔU_{ES} can be obtained by recalculating the electrostatic interaction energy for different dimer pairs using atomic partial charges of the neutral and doped states, sampled over a broad range of configurations, presented in Figure 5. A $1/r$ function

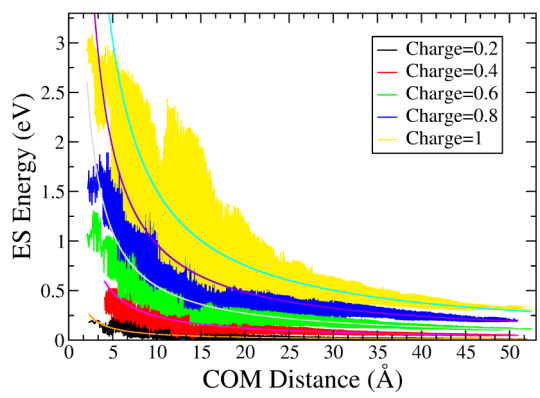

Figure 5. Electrostatic energy distribution for P3HT/F4TCNQ pairs as a function COM separation distances and the amount of partial charge transfer.

has been fit to the energies for each partial charge state, representing the average electrostatic interaction energy between P3HT and F4TCNQ molecules as a function of intermolecular separation between their centers-of-mass (COM). For each partial charge state, we average over all energy points at 5 Å, allowing a width of 0.2 Å.

Figure 6 shows how these fits of the electrostatic energy as a function of the amount of charge transfer can be leveraged, in concert with the linear dependence of charge transfer on the DFT energy, to extract a ΔG^0_{rxn} for partial charge transfer of different CTC states. Usefully, the $\Delta G^0_{rxn}(q)$ function can be fit to a quadratic function, making parametrization straightforward. This parametrization of $\Delta G^0_{rxn}(q)$ is used to run RMCMD incorporating a spectrum of partial charge transfer

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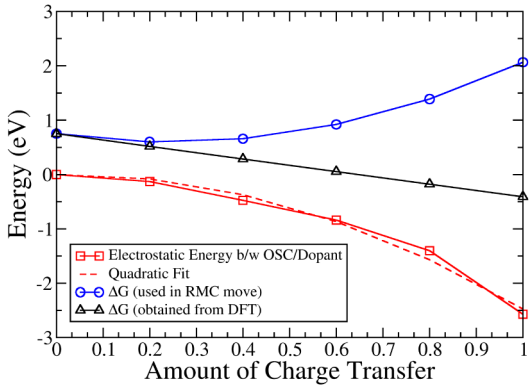

Figure 6. ΔG_{DFT}, $\Delta G^0_{\mathrm{rxn}}$ and ΔU_{ES} plotted as a function of the amount of partial charge transfer, as well as a quadratic fit to $\Delta G^0_{\mathrm{rxn}}$ employed in the RMCMD acceptance criterion.

states ranging between neutral and ICT. As a qualifier, we note that the rigorous determination of the amount of partial charge transfer should rely on calculating the intermolecular electronic coupling between the Highest Occupied Molecular Orbital (HOMO) of P3HT and the Lowest Unoccupied Molecular Orbital (LUMO) of F4TCNQ, but such a calculation is clearly not scalable to the system sizes and sampling required to understand molecular doping.

Figure 7a shows the relative populations of the dopants, defined as

$$\text{Relative Population (at charge state } q) = \frac{\text{Number of dopants at charge state } q}{\text{Total number of dopants}} \tag{14}$$

at the six chosen partial charge transfer states. Interestingly, it is observed that the population of ICT states is dramatically larger than the populations of any of the CTC states, with nearly 80% of the dopants fully ionizing. This trend is consistent with experimental results of P3HT and F4TCNQ showing near unity doping of P3HT, even if not all of the doped states are mobile charge carriers. The strong preference for ICT states over CTC states within RMCMD is due to the stronger electrostatic interaction lowering the $\Delta G^0_{\mathrm{rxn}}$, which results in a larger amount of charge transfer. Figure 7b shows the associated RDF—due to the high proportion of ICTs, this RDF looks similar to Figure 4a.

Provided the reasonable behavior of the RMCMD model for both ICT and CTC states, we next test the ability of our implementation to capture the conformational effects of polaron-induced backbone planarization in the doped state, and its effects on the resulting morphology. For this test, we run simulations in a system of an equal number of P3HT and F4TCNQ molecules to allow the system to fully dope, and thus magnify the effect of backbone planarization on doping. For simplicity, only ICT doping is allowed and $\Delta G^0_{\mathrm{rxn}}$ is set to a value low enough (≈ -4.3 eV) that all molecules in the system ionize. After the system becomes fully doped, RMC is switched off and we perform pure NVT MD for 10 ns. For thiophene ring i in a single P3HT 8-mer backbone, we calculate the vector $\boldsymbol{v}_i$ perpendicular to the plane of the ring. The average planarity $p = \langle p_i \rangle$ indicates the extent of backbone planarization in an OSC molecule, where

$$p_i = \frac{\boldsymbol{v}_i \cdot \boldsymbol{v}_{i-1}}{|\boldsymbol{v}_i||\boldsymbol{v}_{i-1}|} \tag{15}$$

Figure 8 shows the average and standard deviation of the planarity for the 100 OSC molecules in the system. It is clear that the P3HT molecules have become substantially more planar as a result of the doping process, which is the intended effect.

There are many paths moving forward by which to enhance the parametrization of the RMCMD methodology introduced in this work. First, the reaction free energy for partial charge transfer, $\Delta G^0_{\mathrm{rxn}}(q)$, could be straightforwardly fine-tuned to obtain doping efficiencies that quantitatively align with experimental doping efficiencies. In the absence of experimental data, $\Delta G^0_{\mathrm{rxn}}(q)$ could be evaluated for a diverse range of dimer configurations sampled from MD using e.g., Constrained DFT, upon which a surrogate model could be trained and applied within the RMC routine. Second, the polaronic effects within the model could be treated more rigorously, with configuration dependence of the polaronic energy incorporated directly into $\Delta G^0_{\mathrm{rxn}}$. Relatedly, the dihedral coefficients in the doped state, as well as more granular polaronic potential energy terms (e.g., bonds, angles, and other dihedrals) could be incorporated directly into the geometric restructuring after the RMC moves. Presumably, the incorporation of all of these

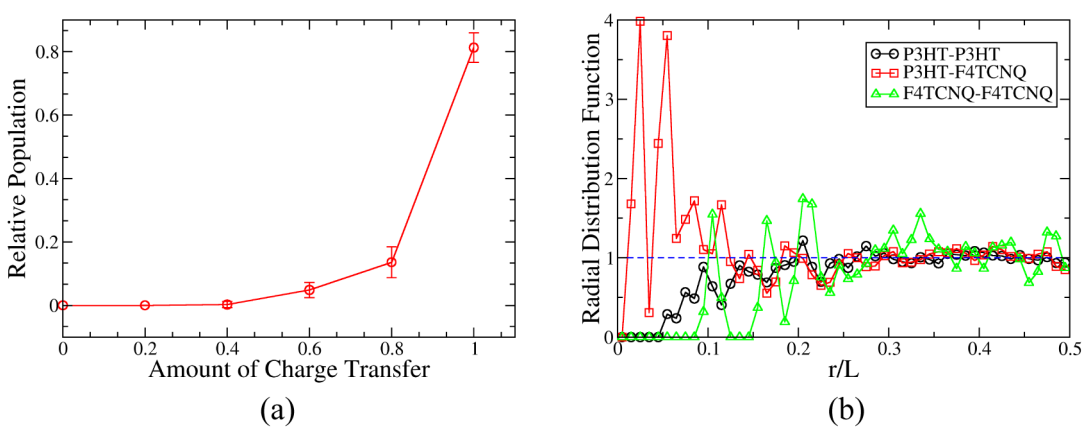

Figure 7. (a) Relative population (as defined in eq 14) as a function of the amount of partial charge transfer, indicating the proportion of CTCs vs ICT. (b) Radial distribution function for a system of 100 P3HT doped with 25 F4TCNQ, allowing for both ICT and CTCs.

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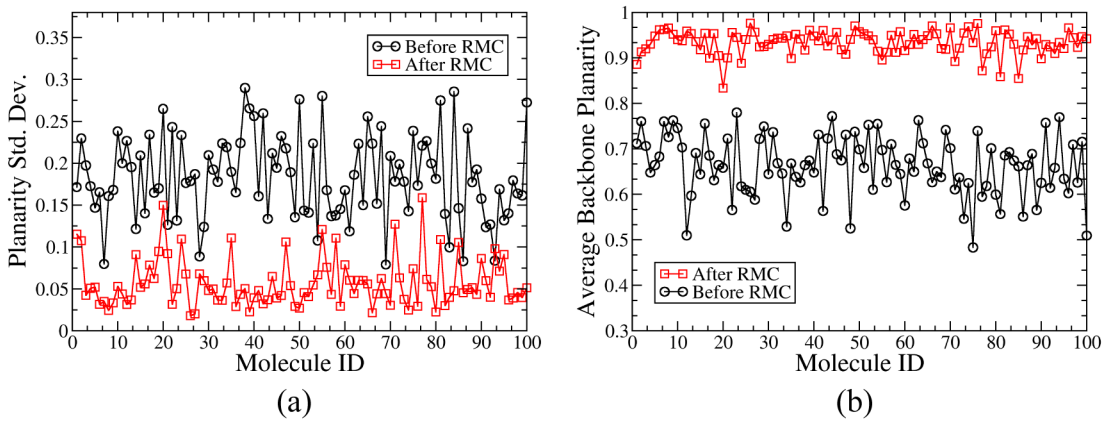

Figure 8. (a) Average planarity and (b) Standard deviation of the planarity for the 100 OSC molecules present in the system, before and after performing RMCMD calculations.

finer geometric motifs upon doping would enhance the accuracy of describing modifications in e.g., π-π stacking. Lastly, one deficiency of the present model is the inability to incorporate delocalization of charge carrier across multiple OSC or dopant molecules, which is a feature observed in experimental works. Incorporating delocalization within a classical RMCMD model is an extremely interesting technical challenge, as it requires modifications to the RMC move set as well as robust and efficient parametrizations of ΔG^0_{rxn} for different delocalized states. These topics all constitute directions for future work.

5. CONCLUSION

We have introduced the all-atom RMCMD method for the computational study of molecular doping in organic semiconductors. This approach combines classical MD with MC moves for the doping reaction, enabling simultaneous study of the morphology and doping efficiency of doped OSCs. Atomistic charges and reaction free-energies for the neutral and doped states are obtained from first-principles calculations, allowing the method to capture ICTs, CTCs and the backbone planarization associated with polaron formation. We applied this method to P3HT doped with F4TCNQ and observed qualitative agreement with previous experimental and first-principles work,[17,20,29] Without RMCMD, such studies are not practical since classical MD alone does not provide insight into electronic behavior and first-principles electronic structure methods do not scale to the characteristic length scales of doped OSC systems. There are many exciting applications of our method—we can compare doping efficiencies across different dopants and OSCs, gauge the effect of dopant aggregation and can even capture the effects of explicit solvent molecules in the simulations. These applications are the subject of ongoing work.

■ ASSOCIATED CONTENT

Data Availability Statement

The RMCMD LAMMPS fix is made available on GitHub.[51] LAMMPS input files, structure files, RMCMD specific input files, as well as output structures and log files are made available on Box.[62]

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.5c01206.

Histograms representing the electrostatic self-energy distributions for the semiconductor and dopant molecules, with a short discussion about the distribution widths; Additional details about the partial charge free energy profile, with a plot depicting the linear trend between the extent of charge transfer and the reaction free energy ΔG_{rx} obtained from first-principles calculations; Histogram representing the distribution of polaronic energy differences, with additional details on utilizing the distribution to tune relevant dihedral parameters (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Office of Naval Research under Grant No. N00014-23-1-2542. The RMCMD simulations were run on the Purdue Anvil supercomputer using Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) allocation CHE250054. A.V. acknowledges that this material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE 21-46756.

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